

AN APPROXIMATION TO THE DISTRIBUTION OF HOTELLING'S GENERALIZED T_0^2 -STATISTIC

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Summary: An approximation to the distribution of Hotelling's Generalized T_0^2 in terms of Snedecor's F-distribution is presented for the central as well as the noncentral case. This is done by ignoring higher order terms in the asymptotic expansion of the distribution function.

1. INTRODUCTION

Let S_e and S_h be independent Wishart matrices: $S_e \sim W_p(\Sigma, n)$ and $S_h \sim W_p(\Sigma, m, \Omega)$.

The T_0^2 -statistic is defined by (see e.g. Pillai and Young (1971)):

$$(1.1) \quad T_0^2 = nT = n\text{tr}(S_h S_e^{-1}).$$

The T_0^2 -statistic is a criterion for the multivariate linear hypothesis, equivalent to $\Omega = 0$. The central distribution ($\Omega = 0$) of T_0^2 is the same as that of other statistics, e.g. that of a test for equality of covariances. Constantine (1966) derived the distribution of T in terms of zonal polynomials. This expansion is valid for $|T| < 1$ only. Davis (1968) showed that the density of T satisfies a linear differential equation of order p . Davis (1970a to 1980) computed percentage points for the central distribution of T by analytic continuation of Constantine's (1966) expansion. Siotani (1971) derived an asymptotic expansion to order n^{-2} for the distribution of T_0^2 and proved that, for $p = 2$, the terms of order n^{-2} contributed significantly to the calculation of the power function. Krishnaiah and Chang (1972) derived the exact central distribution of T as linear combinations of inverse Laplace transforms of certain double integrals. Hayakawa (1972) used Laguerre polynomials to find an asymptotic expansion for the probability density function of T_0^2 , which is valid for $0 \leq T_0^2 \leq n$, and includes the term of order n^{-2} . Fujikoshi (1978) used a perturbation method to derive an asymptotic expansion of the distribution of T . This expansion is based on the assumption that $\Omega = O(n)$. Chattopadhyay (1975) derived an asymptotic expansion, to terms of order n^{-1} , for the case where the expected values of S_h and S_e differ. This expansion is a generalization of Siotani's (1971) result. None of these expansions used an F-distribution as first term.

The expansion derived in section 2 of this paper specifies an F-distribution as first term. The parameters of the F-distribution are chosen in such a way as to minimize the remaining terms of the series.

Pillai and Young (1971) compared, among others, three approximations to the central distribution of T . These approximations consisted of fitting the first, the first two and the first three moments of a type F (Pearson type II) distribution, respectively. Davis (1970b) also suggested a Pearson curve approximation. Hughes and Saw (1972) proposed two different approximations: based on an F -distribution when $m+n > mp+1$, and on a χ^2 -distribution otherwise.

The asymptotic expansion derived in this paper leads to the suggestion, in section 3, that the central distribution of T be approximated as follows:

$$(1.2) \quad F_T(\xi) \approx F(q\xi)$$

where $F(\cdot)$ is the central $F(mp, \beta)$ distribution function, with

$$(1.3) \quad \beta = (m+n)(mp+2)/(m+p+1) - mp.$$

$$q = (m+n)/mp - (m+p+1)/(mp+2).$$

The error of approximation is of order $(m+n)^{-2}$.

When $m = 1$, (1.2) reduces to the known exact distribution. Percentage points calculated with (1.2) compare well with published exact values. The approximation is good, provided $q > 0.1$ which is usually the case in practice and is especially accurate in the tails of the distribution. Hence it is applicable for a wide range of p, m and n values as indicated numerically in section 3.

In section 4 an approximation to the noncentral distribution of T is derived. The distribution function of T is approximated with

$$(1.4) \quad F_T(\xi) \approx F(q\xi) + \mu [g_{mp+6}(\rho\xi) - g_{mp+8}(\rho\xi)]$$

where $g_{\alpha}(\cdot)$ is the noncentral $\chi^2(\alpha, \lambda)$ p.d.f. and $F(\cdot)$ the noncentral $F(mp, \beta, \lambda)$ distribution function. The values of q and β are given in (1.3) and

$$(1.5) \quad \lambda = \text{tr}(\Omega)$$

$$\rho = m + n$$

$$\mu = \frac{1}{2\rho} \left[\text{tr}(\Omega^2) - \frac{m+p+1}{mp+2} \lambda^2 \right]$$

The approximation ignores terms of order $(m+n)^{-2}$. It leads to the conclusion that the magnitude of the differences among the eigenvalues of Ω have an interesting effect on the power function of the test.

2. ASYMPTOTIC EXPANSION

We assume that S_e has a nonsingular distribution. If S_h is singular the distribution of T may be found by exchanging m with p , n with $n+m-p$, and p with m . When $\Omega = 0$, the distribution of T is independent of Σ , and when $\Omega \neq 0$, the distribution is a function of $\Sigma^{-1}\Omega$ only. Hence we may assume, without loss of generality, that $\Sigma = I$ and $n > m \geq p$. Let S_h and S_e be independent Wishart matrices, with distributions $W_p(I, m, \Omega)$ and $W_p(I, n)$ respectively. It is convenient to consider the statistic

$$(2.1) \quad \underline{w} = \frac{\rho}{n} T_0^2 = \rho \bar{T} = \rho \text{tr}(S_h S_e^{-1})$$

where ρ is an arbitrary constant.

The characteristic function of $\underline{w}$ is:

$$(2.2) \quad \phi(t) = E_{\tilde{W}}(e^{it\tilde{W}}) = E_{S_e, S_h}(e^{it\rho \operatorname{tr}(S_h S_e^{-1})}).$$

It will be shown that

$$(2.3) \quad \phi(t) = c\phi_0(t) + e^{it\lambda(1-2it)^{-1}} \cdot \sum_{k=2}^{\infty} \mu_k (1-2it)^{-\frac{1}{2}mp-k}$$

where $\lambda = \operatorname{tr} \Omega$ and $\phi_0(t)$ is the characteristic function of $\rho_1(\alpha|\beta) \cdot F(\alpha, \beta, \lambda)$. The constants ρ_1 , α and β will be specified. $\mu_2 = O((m+n)^{-1})$. If $\lambda = 0$, then $\mu_k = O((m+n)^{-[\frac{1}{2}(k+1)])}$ and when $\lambda \neq 0$, $\mu_k = O((m+n)^{-[\frac{1}{2}(k-1)])}$ for $k = 3, 4, 5, \dots$, where $[a]$ denotes the integer part of a . Because $S_h \sim W(I, m, \Omega)$, independent of S_e , we have, from (2.2):

$$(2.4) \quad \phi(t) = E_{S_e} \{ |I - 2it\rho S_e^{-1}|^{-\frac{1}{2}m} \cdot \operatorname{etr}(it\rho S_e^{-1}(I - 2it\rho S_e^{-1})^{-1}\Omega) \}.$$

Using the fact that:

$$(2.5) \quad (I - 2it\rho S_e^{-1})^{-1} = \rho^{-1}(1-2it)^{-1} S_e (I - (1-2it)^{-1}(I - \rho^{-1}S_e))^{-1}$$

and putting $\theta = \theta(t) = (1-2it)^{-1} \therefore it(1-2it)^{-1} = \frac{1}{2}(\theta-1)$, it follows that:

$$\begin{aligned} (2.6) \quad \phi(t) &= E_{S_e} \{ (\theta/\rho)^{\frac{1}{2}mp} |S_e|^{\frac{1}{2}m} |I - \theta(I - \rho^{-1}S_e)|^{-\frac{1}{2}m} \operatorname{etr}(\frac{1}{2}(\theta-1)(I - \theta(I - \rho^{-1}S_e))^{-1}\Omega) \} \\ &= E_S \{ \Gamma_p(\frac{1}{2}(m+n)) / \Gamma_p(\frac{1}{2}n) 2^{\frac{1}{2}mp} (\theta/\rho)^{\frac{1}{2}mp} |I - \theta(I - \rho^{-1}S)|^{-\frac{1}{2}m} \\ &\quad \cdot \operatorname{etr}(\frac{1}{2}(\theta-1)(I - \theta(I - \rho^{-1}S))^{-1}\Omega) \} \end{aligned}$$

where $S \sim W(I, m+n)$.

But:

$$\begin{aligned}
 (2.7) \quad & \text{tr}\{\frac{1}{2}(\theta-1)(I-\theta(I-\rho^{-1}S))^{-1}\Omega\} \\
 &= \text{tr}\{\frac{1}{2}(\theta-1)[I-\theta(I-\rho^{-1}S) + \theta(I-\rho^{-1}S)(I-\theta(I-\rho^{-1}S))^{-1}]\Omega\} \\
 &= \text{tr}(\frac{1}{2}(\theta-1)\Omega) + \text{tr}\{\frac{1}{2}\theta(\theta-1)(I-\rho^{-1}S)(I-\theta(I-\rho^{-1}S))^{-1}\Omega\} \\
 &= \text{tr}(\frac{1}{2}(\theta-1)\Omega) - \frac{1}{2}\theta \text{tr}(I-\rho^{-1}S)\Omega + \sum_{j=2}^{\infty} \theta^j (\text{tr}(I-\rho^{-1}S)^{j-1}\Omega - \text{tr}(I-\rho^{-1}S)^j\Omega).
 \end{aligned}$$

Let:

$$(2.8) \quad c_1 = (2/\rho)^{\frac{1}{2}mp} \Gamma_p(\frac{1}{2}(m+n)) / \Gamma_p(\frac{1}{2}n).$$

Then, from (2.6) and (2.7):

$$\begin{aligned}
 (2.9) \quad \phi(t) &= c_1 e^{\frac{1}{2}(\theta-1)\text{tr}(\Omega)\theta^{\frac{1}{2}mp}} E_S[\exp(\sum_{j=1}^{\infty} \kappa_j(S)\theta^j)] \\
 &= c_1 e^{\frac{1}{2}(\theta-1)\text{tr}(\Omega)} E_S[\sum_{k=0}^{\infty} \eta_k(S)\theta^{\frac{1}{2}mp+k}]
 \end{aligned}$$

where

$$\begin{aligned}
 (2.10) \quad \kappa_1(S) &= \frac{1}{2}[\text{tr}(I-\rho^{-1}S)(mI-\Omega)] \\
 \kappa_j(S) &= \frac{1}{2j}[\text{tr}(I-\rho^{-1}S)^j(mI-j\Omega) + j \cdot \text{tr}((I-\rho^{-1}S)^{j-1}\Omega)]
 \end{aligned}$$

and

$$(2.11) \quad \eta_0(S) = 1$$

$$\eta_k(S) = \frac{1}{k} \sum_{r=0}^{k-1} (k-r) \kappa_{k-r}(S) \eta_r(S).$$

The expansions derived by Davis (1970a) (central case), Siotani (1971) and Hayakawa (1972) are special cases of (2.9), with $\rho = n$.

We have specifically:

$$(2.12) \quad E_S(\eta_1(S)) = E_S(\frac{1}{2} \text{tr}(I - \rho^{-1}S)(mI - \Omega)) \\ = \frac{1}{2}(1 - \rho^{-1}(m+n))(\text{mp} - \text{tr}(\Omega)) .$$

If ρ is chosen to be

$$(2.13) \quad \rho = m+n$$

then the coefficient of θ in (2.9) will vanish, that is:

$$(2.14) \quad E_S(\eta_1(S)) = 0 .$$

Also, with this choice of ρ , the random matrix

$$(2.15) \quad \frac{S - \rho I}{\sqrt{\rho}} = \sqrt{\rho}(\rho^{-1}S - I)$$

will converge in distribution to a normal distribution, which is independent of ρ . Hence the expected value of a homogeneous polynomial of degree k in the elements of $(I - \rho^{-1}S)$ will be $O(\rho^{-\frac{1}{2}k})$.

The order of $E_S(\text{tr}(I - \rho^{-1}S)^j)$ is an integer, hence

$$(2.16) \quad E_S(\text{tr}(I - \rho^{-1}S)^j) = O(\rho^{-[\frac{1}{2}(j+1)]}) .$$

From (2.11) it now follows that, if $\Omega = 0$, then

$$(2.17) \quad E_S(\eta_k(S)) = O(\rho^{-[\frac{1}{2}(k+1)]}) \quad k = 2, 3, 4, \dots$$

and when $\Omega \neq 0$:

$$(2.18) \quad E_S(\eta_k(S)) = O(\rho^{-1}) \quad k = 2 \\ = O(\rho^{-[\frac{1}{2}(k-1)]}) \quad k = 3, 4, 5, \dots$$

Specifically:

$$\begin{aligned}
 (2.19) \quad E_S(\eta_2(S)) &= E_S[\tfrac{1}{2}[2\kappa_2(S) + \eta_1(S)\kappa_1(S)]] \\
 &= E_S[\tfrac{1}{2}(\text{tr}(I - \rho^{-1}S)^2(mI - 2\Omega) + 2\text{tr}(I - \rho^{-1}S)\Omega) \\
 &\quad + \tfrac{1}{2}(\tfrac{1}{2}\text{tr}(I - \rho^{-1}S)(mI - \Omega))^2] \\
 &= \tfrac{1}{2}\rho^{-1}(p+1)\text{tr}(mI - 2\Omega) + 0 + \tfrac{1}{8}[2\rho^{-1}\text{tr}(mI - \Omega)^2] \\
 &= \tfrac{1}{2}\rho^{-1}[(mp - 2\text{tr}\Omega)(m+p+1) + \text{tr}(\Omega^2)] .
 \end{aligned}$$

The characteristic function of $\rho_1(\alpha/\beta) \cdot \underline{z}$, where $\underline{z} \sim F(\alpha, \beta, \lambda)$ may be expanded similarly namely

$$(2.20) \quad \phi_0(t) = c_2 e^{\frac{1}{2}(\theta-1)\lambda} E_{\underline{u}} \left[\sum_{k=0}^{\infty} \eta'_k(\underline{u}) \theta^{\frac{1}{2}\alpha+k} \right]$$

where $\underline{u} \sim \chi^2(\alpha+\beta)$ and

$$(2.21) \quad c_2 = (2/\rho_1)^{\frac{1}{2}\alpha} \Gamma(\tfrac{1}{2}(\alpha+\beta)) / \Gamma(\tfrac{1}{2}\alpha) .$$

Also

$$\begin{aligned}
 (2.22) \quad \eta'_0(\underline{u}) &= 1 \\
 \eta'_k(\underline{u}) &= \frac{1}{k} \sum_{r=0}^{k-1} (k-r) \kappa'_{k-r}(\underline{u}) \eta'_r(\underline{u})
 \end{aligned}$$

where

$$\begin{aligned}
 (2.23) \quad \kappa'_1(\underline{u}) &= \tfrac{1}{2}(1 - \rho_1^{-1}\underline{u})(\alpha - \lambda) \\
 \kappa'_r(\underline{u}) &= \frac{1}{2^j} [(1 - \rho_1^{-1}\underline{u})^j (\alpha - j\lambda) + j\lambda(1 - \rho_1^{-1}\underline{u})^{j-1}]
 \end{aligned}$$

specifically

$$(2.24) \quad E_{\underline{u}}(\eta'_1(\underline{u})) = (1 - \rho_1^{-1}(\alpha+\beta))(\alpha - \lambda) .$$

A convenient choice for the parameter β follows:

$$(2.25) \quad \rho_1 = \alpha + \beta \therefore \beta = \rho_1 - \alpha.$$

Hence, the coefficient of θ in (2.20) becomes

$$(2.26) \quad E_{\underline{u}}(\eta_1'(\underline{u})) = 0.$$

Also

$$\begin{aligned} (2.27) \quad E_{\underline{u}}(\eta_2'(\underline{u})) &= \frac{1}{2} E_{\underline{u}}[(1 - \rho_1^{-1} \underline{u})^2 (\alpha - 2\lambda) + 2\lambda(1 - \rho_1^{-1} \underline{u}) \\ &\quad + [(1 - \rho_1^{-1} \underline{u})(\alpha - \lambda)]^2] \\ &= \frac{1}{2} \rho_1^{-1} [(\alpha + 2)(\alpha - 2\lambda) + \lambda^2]. \end{aligned}$$

Comparing (2.20) and (2.9) indicates the following choices:

$$(2.28) \quad \alpha = mp$$

$$(2.29) \quad \lambda = \text{tr}(\Omega).$$

Hence

$$(2.30) \quad \phi(t) - (c_1/c_2)\phi_0(t) = e^{\frac{1}{2}(\theta-1)\lambda} \sum_{k=2}^{\infty} \mu_k \theta^{\frac{1}{2}mp+k}$$

where

$$(2.31) \quad \mu_k = c_1 \{E_S(\eta_k(S)) - E_{\underline{u}}(\eta_k'(\underline{u}))\}.$$

Now (2.30) is in the form (2.3) with $c = c_1/c_2$. The constant ρ_1 may now be chosen. From (2.19) and (2.27)

$$(2.32) \quad \mu_2 = \frac{1}{2} c_1 [\rho^{-1}((mp+1)(mp-2\lambda) + \text{tr}(\Omega^2)) - \rho_1^{-1}((mp+2)(mp-2\lambda) + \lambda^2)].$$

A possible choice for ρ_1 is:

$$(2.33) \quad \rho_1 = \alpha + \beta = \rho(mp+2)/(m+p+1) \\ = (m+n)(mp+2)/(m+p+1) .$$

With this choice, we have:

$$(2.34) \quad \mu_2 = \frac{1}{2}c_1\rho^{-1}[\text{tr}(\Omega^2) - \frac{m+p+1}{mp+2}\chi^2] \\ = 0 \quad \text{for } \Omega = 0$$

$$(2.35) \quad \mu_3 = -\frac{1}{2}c_1\rho^{-1}[\text{tr}(\Omega^2) - \frac{m+p+1}{mp+2}\chi^2] + O(\rho^{-2}) \\ = -2\mu_2 + O(\rho^{-2})$$

$$(2.36) \quad \mu_4 = \frac{1}{2}c_1\rho^{-1}[\text{tr}(\Omega^2) - \frac{m+p+1}{mp+2}\chi^2] + O(\rho^{-2}) \\ = \mu_2 + O(\rho^{-2}) .$$

Generally, for $k=5,6,7,\dots$ (similar to (4.2.17) and (4.2.18), we have

$$(2.37) \quad \mu_k = O(\rho^{-[\frac{1}{2}(k+1)]}) \quad \text{for } \Omega = 0 \\ = O(\rho^{-[\frac{1}{2}(k-1)]}) \quad \text{for } \Omega \neq 0 .$$

Putting $t=0$ in (2.30) yields

$$(2.38) \quad \frac{c_1}{c_2} = 1 - \sum_{k=2}^{\infty} \mu_k .$$

The characteristic function of $w = \rho T$ is therefore

$$\begin{aligned}
 (2.39) \quad \phi(t) &= (c_1/c_2)\phi_0(t) + \sum_{k=2}^{\infty} \mu_k \phi_{\tilde{x}_k}(t) \\
 &= \phi_0(t) + \sum_{k=2}^{\infty} \mu_k (\phi_{\tilde{x}_k}(t) - \phi_0(t)) \\
 &= \phi_0(t) + \mu_2 [\phi_{\tilde{x}_2}(t) - 2\phi_{\tilde{x}_3}(t) + \phi_{\tilde{x}_4}(t)] + O(\rho^{-2})
 \end{aligned}$$

where

$$\begin{aligned}
 (2.40) \quad \phi_{\tilde{x}_k}(t) &= e^{\frac{1}{2}(\theta-1)\lambda_{\theta} \frac{1}{2}mp+k} \\
 &= e^{it\lambda(1-2it)^{-1}} \cdot (1-2it)^{-\frac{1}{2}mp-k}
 \end{aligned}$$

which is the characteristic function of $\chi^2(mp+2k, \lambda)$. Hence (2.39) is the same as (2.3).

Inverting each term of (2.39) yields an expression for the distribution function of $\underline{w}$.

$$(2.41) \quad F_{\underline{w}}(w) = F_0(w) + \sum_{k=2}^{\infty} \mu_k [G_{mp+2k}(w) - F_0(w)].$$

The distribution function of $T = \underline{w}/\rho$ follows directly:

$$(2.42) \quad F_T(\xi) = F(q\xi) + \sum_{k=2}^{\infty} \mu_k [G_{mp+2k}(\rho\xi) - F(q\xi)]$$

$F_0(\cdot)$ is the distribution function of $\rho_1(\alpha/\beta) \cdot F(\alpha, \beta, \lambda)$,

$F(\cdot)$ is the $F(\alpha, \beta, \lambda)$ distribution function and

$G(\cdot)$ is a $\chi^2(\alpha, \lambda)$ distribution function, with

$$(2.43) \quad \alpha = mp$$

$$\rho = m+n$$

$$\rho_1 = \rho(mp+2)/(m+p+1)$$

$$\beta = \rho_1 - \alpha$$

$$\lambda = \text{tr}(\Omega)$$

$$q = \rho\beta/(\rho_1\alpha) = (m+n)/mp - (m+p+1)/(mp+2)$$

and μ_k is given by (2.34), (2.35), (2.36) and (2.31).

3. THE CENTRAL CASE

In the central case ($\Omega = 0$), (2.42) reduces to:

$$(3.1) \quad F_T(\xi) = F(q\xi) + \sum_{k=3}^{\infty} \mu_k [G_{mp+k}(\rho\xi) - F(q\xi)] \text{ because } \mu_2 = 0.$$

$F(\cdot)$ and $G_{mp+k}(\cdot)$ are central distribution functions ($\lambda = 0$), with parameters given by (2.43) and μ_k is given by (2.31).

It was found numerically, that the terms of the series on the RHS of (3.1) contributed minimally to the value of the distribution function of T . The terms are of order $(m+n)^{-2}$. It follows that

$$(3.2) \quad T \text{ is distributed approximately as } q^{-1} \cdot F(mp, \beta) \text{ with}$$

$$q = (m+n)/mp - (m+p+1)/(mp+2)$$

$$\beta = (m+n)(mp+2)/(m+p+1) - mp.$$

When $q < 0,1$ the approximation is poor, see e.g. 4; 4; 5 ($q = 0,0625$) and 10; 10; 14 ($q = 0,034$) in table 2.

Hughes and Saw (1972) suggested that:

(3.3) T is distributed approximately as $\gamma_1 \cdot F(\nu_1, \nu_2)$ provided $\nu_1 > 0$ and

(3.4) T is distributed approximately as $\gamma_2 \cdot \chi^2(\nu_3)$, otherwise, with

$$\gamma_1 = mp/(n-p+1)$$

$$\nu_1 = mp(n-p)/(n+m-pm-1)$$

$$\nu_2 = n-p+1$$

$$\gamma_2 = (n-1)(m+n-p-1)/\{(n-p)(n-p-1)(n-p-3)\}$$

$$\nu_3 = mp(n-p)(n-p-3)/\{(n-1)(m+n-p-1)\}$$

When $\nu_1 < 0$ and $n-p \leq 3$ neither approximation can be used, because $\nu_3 < 0$.

SAS Institute Inc. (1982) uses the following approximation:

(3.5) T is distributed approximately as $\gamma_3 \cdot F(\nu_4, \nu_5)$, with

$$\nu_4 = mp$$

$$\nu_5 = p \cdot (n-p-1) + 2$$

$$\gamma_3 = p \cdot \nu_4 / \nu_5$$

The BMDP-83 system uses Tiku (1971)'s approximation:

(3.6) T is distributed as $\gamma_4 \cdot \chi^2(\nu_6) - \gamma_5$ provided $n > 30+p+1$ and

(3.7) T is distributed as $\gamma_6/\tilde{x} - \gamma_7$ where $\tilde{x} \sim \chi^2(\nu_7)$, otherwise with

$$\gamma_4 = \frac{(2m+n-p-1)(n+p-1)}{(n-p-5)(n-p-1)(n-p+1)}$$

$$\nu_6 = \frac{mp(m+n-p-1)(n-p+1)^2(n-p-5)^2(n-1)}{(2m+n-p-1)^2(n+p-1)^2(n-p)(n-p-3)}$$

$$\gamma_5 = \gamma_4 \cdot \nu_6 - mp/(n-p-1)$$

$$\nu_7 = 6 + 2 \cdot \nu_6 \cdot \{1 + \sqrt{1 + 2/\nu_6}\}$$

$$\gamma_6 = \frac{1}{2}\gamma_4(\nu_7-2)(\nu_7-6)$$

$$\gamma_7 = \frac{1}{2}\gamma_4(\nu_7-6) - mp/(n-p-1).$$

In tables 1 and 2, comparisons of the various approximations mentioned, with the exact results, are given. The exact values were taken from Pillai and Jayachandran (1967) for table 1 and from Davis (1970a, 1970b and 1980) for table 2. The table headings refer to the following approximations:

<u>Heading</u>	<u>Approximation</u>
A	(3.2)
H and S	(3.3) or (3.4) with *
SAS	(3.5)
BMDP	(3.6) or (3.7) with *

It is clear that (3.2) is an improvement over the others. It is always valid, and it yields exact values for smaller n . The percentage points for 10; 10; 14 and 4; 4; 5 appear to be very inaccurate, but the corresponding p -values will obviously be quite close to the exact probabilities.

TABLE 1
 Percentage points of $T = \text{tr}(S_h S_e^{-1})$ for $p=2$

m	n	5% points					1% points				
		BMDP	SAS	H and S	A	Exact	BMDP	SAS	H and S	A	Exact
3	13	1,442	1,390	1,445	1,447	1,451	2,263	2,050	2,294	2,232	2,270
	33	0,437	0,435	0,443	0,443	0,443	0,621	0,602	0,628	0,622	0,624
	53	0,262	0,258	0,260	0,260	0,260	0,359	0,351	0,361	0,358	0,359
	83	0,161	0,160	0,161	0,161	0,161	0,219	0,216	0,220	0,219	0,219
5	13	2,176	2,088	2,170	2,162	2,177	3,278	2,961	3,319	3,170	3,264
	53	0,383	0,377	0,382	0,381	0,381	0,500	0,490	0,504	0,499	0,500
	83	0,236	0,233	0,235	0,235	0,235	0,304	0,300	0,306	0,304	0,304
13	13	4,962	4,755	4,994*	4,857	4,941	7,084	6,424	6,443*	6,674	7,020
	83	0,506	0,502	0,506	0,504	0,505	0,606	0,602	0,612	0,606	0,607

TABLE 2

Percentage points of $\frac{n}{m}T = \frac{1}{m}T_0^2 = \frac{n}{m}\text{tr}(S_h S_e^{-1})$

p	m	n	5% points					1% points					
			BMDP	SAS	H and S	A	Exact	BMDP	SAS	H and S	A	Exact	
3	3	6	-	22,870	-	25,591	25,456	-	39,897	-	53,653	59,507	
		7	-	16,588	26,887*	18,580	18,752	-	26,526	46,445*	34,498	37,994	
		40	6,645	6,434	6,591	6,581	6,588	8,822	8,420	8,908	8,778	8,811	
		100	6,005	5,935	5,991	5,988	5,989	7,790	7,662	7,832	7,788	7,793	
3	4	7	-	15,965	25,653	17,702	18,056	-	25,185	43,164	31,616	35,993	
		6	-	21,420	-	23,374	23,899	-	36,529	-	44,704	54,377	
		8	-	12,412	16,476*	13,431	13,934	-	18,289	24,459*	21,393	24,755	
		80	5,223	5,162	5,222	5,209	5,212	6,405	6,303	6,478	6,401	6,412	
4	4	5	-	777,33	-	388,666	199,640	-	3977,500	-	1988,734	999,920	
		5	9,475	9,129	9,471	9,393	9,429	11,578	10,951	11,883	11,482	11,593	
		5	8,660	8,639	8,666	8,659	8,659	9,978	9,943	10,007	9,979	9,980	
		6	8,579	8,569	8,582	8,579	8,579	9,873	9,856	9,887	9,874	9,874	
6	6	500	11,413*	10,905	11,323*	11,246	11,395	13,600*	12,573	13,124*	13,204	13,586	
		1000	9,877	9,592	9,800*	9,769	9,854	11,278	10,794	11,051*	11,104	11,298	
		10	8,336	8,260	8,374	8,321	8,333	9,269	9,147	9,437	9,252	9,276	
		25	100	7,965	7,905	7,947*	7,950	7,963	8,698	8,604	8,649*	8,677	8,703
		35	40	9,286	9,069	9,218*	9,163	9,270	10,345	9,983	10,146*	10,141	10,361
		100	7,779	7,726	7,762*	7,762	7,777	8,415	8,334	8,367*	8,391	8,419	
		500	7,140	7,133	7,150	7,140	7,140	7,624	7,613	7,656	7,624	7,625	
		1000	7,067	7,063	7,072	7,067	7,067	7,534	7,528	7,550	7,534	7,534	
		10	10	-	73,004	131,857*	83,720	98,999	-	91,094	198,013*	115,082	180,900
		40	18,586*	17,769	18,463*	18,355	18,569	21,271*	19,771	20,765*	20,784	21,267	
		100	14,343	14,142	14,418	14,316	14,338	15,871	15,544	16,187	15,835	15,881	
		35	10	500	12,774	12,743	12,789	12,773	12,774	13,984	13,934	14,039	13,984
1000	12,602			12,587	12,609	12,602	12,602	13,780	13,755	13,806	13,780	13,780	
70	14,006			13,809	13,966*	13,927	14,002	15,036	14,724	14,926*	14,910	15,046	
100	13,063			12,949	13,041*	13,028	13,061	13,919	13,741	13,857*	13,865	13,925	
500	11,594			11,578	11,612	11,593	11,594	12,209	12,185	12,266	12,208	12,210	
1000	11,433			11,425	11,442	11,432	11,432	12,023	12,011	12,051	12,023	12,023	

4. THE POWER FUNCTION

The characteristic function of $\underline{w} = \rho T$ is given in (2.39).

$$(4.1) \quad \phi(t) = \phi_0(t) + \mu_2[\phi_{\underline{x}_2}(t) - 2\phi_{\underline{x}_3}(t) + \phi_{\underline{x}_4}(t)] + O(\rho^{-2}).$$

From (2.40) follows that

$$(4.2) \quad \phi_{\underline{x}_k}(t) - \phi_{\underline{x}_{k+1}}(t) = -2it\phi_{\underline{x}_{k+1}}(t).$$

Hence the characteristic function of $\underline{w}$ may be presented as:

$$(4.3) \quad \phi(t) = \phi_0(t) + 2\mu_2[-it\phi_{\underline{x}_3}(t) + it\phi_{\underline{x}_4}(t)] + O(\rho^{-2}).$$

Inverting the LHS of (4.2), and partial integration of the resulting expression, yields

$$(4.4) \quad \int_{-\infty}^{\xi} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} (-2it)\phi_{\underline{x}_{k+1}}(t) dt dx = G_{mp+2k}(\xi) - G_{mp+2(k+1)}(\xi) \\ = 2g_{mp+2(k+1)}(\xi).$$

The distribution function of $\underline{w}$ may be approximated by inverting each term of (4.3) hence

$$(4.5) \quad F_{\underline{w}}(w) \approx F_0(w) + 2\mu_2[g_{mp+6}(w) - g_{mp+8}(w)]$$

$F_0(\cdot)$ corresponds to $\phi_0(t)$ (see (2.20)).

The distribution function of T follows directly from (4.5), namely

$$(4.6) \quad F_T(\xi) \approx F(q\xi) + 2\mu_2[g_{mp+6}(\rho\xi) - g_{mp+8}(\rho\xi)].$$

$F(\cdot)$ is the distribution function of $F(\alpha, \beta, \lambda)$ and $G_\alpha(\cdot)$ and $g_\alpha(\cdot)$ are the distribution function and p.d.f. of $\chi^2(\alpha, \lambda)$, respectively. The parameters are given by (2.43).

The power of T at a percentage point ξ , may (from (4.6)) be approximated by

$$(4.7) \quad P(T > \xi) \approx (1 - F(q\xi)) + 2\mu_2[g_{mp+8}(\rho\xi) - g_{mp+6}(\rho\xi)]$$

The error of the approximation is $O(\rho^{-2})$.

If n is large relative to p and m , (4.7) yields a good approximation to the power of T . For calculation purposes, use is made of the fact that (see (2.8))

$$(4.8) \quad c_1 = 1 + O((m+n)^{-1})$$

Hence (see (2.34))

$$(4.9) \quad \mu_2 = \frac{1}{4\rho}[\text{tr}(\Omega^2) - \frac{m+p+1}{mp+2}\lambda^2] + O(\rho^{-2})$$

In table 3, the approximate power of T , calculated with (4.7), for $p=2$ is presented, together with the corresponding exact values, as given by Pillai and Jayachandran (1967). λ_1 and λ_2 are the eigenvalues of Ω .

It is interesting to note that the first term on the RHS of (4.7) is, as far as Ω is concerned, a function of $\lambda = \text{tr}(\Omega)$ only. The last term, however, which is

$$(4.10) \quad \frac{c_1}{2\rho} \left\{ \text{tr}(\Omega^2) - \frac{m+p+1}{mp+2}\lambda^2 \right\} (g_{mp+8}(\rho\xi) - g_{mp+6}(\rho\xi))$$

does yield a value which depends on the differences among the eigenvalues of Ω . For a fixed value of $\lambda = \text{tr}(\Omega)$, $\text{tr}(\Omega^2)$

TABLE 3

Comparison of power of $T = \text{tr}(S_n S_e^{-1})$ at 5% level for $p = 2$

m	n	λ_1	λ_2	P and J	Approximation
3	33	0,25	0,25	0,0676	0,0675
		0,5	0	0,0676	0,0675
		1	0	0,0871	0,0871
		2	0	0,1311	0,1313
		3	0	0,1807	0,1810
3	63	10	0	0,603	0,6034
		5	5	0,605	0,6064
3	83	0,25	0,25	0,0693	0,0693
		0,5	0	0,0693	0,0693
		1	0	0,0909	0,0909
		3	0	0,1952	0,1953
5	33	3	0	0,1420	0,1424
5	63	10	0	0,492	0,4929
		5	5	0,493	0,4948
13	63	10	0	0,292	0,2942
		5	5	0,292	0,2940
	83	1	0	0,0658	0,0654
		2	0	0,0838	0,0839
		1	1	-	0,0839
		3	0	0,1044	0,1043

will be a minimum when all the eigenvalues are equal. This minimum is λ^2/p . The maximum value of $\text{tr}(\Omega^2)$, for $\lambda = \text{tr}(\Omega)$ fixed, is obtained when λ is the only nonzero eigenvalue of Ω ($\therefore$ the linear case). This maximum is λ^2 .

Also, if x_0 is the mode of $\chi^2(mp+8, \lambda)$ then

$$\begin{aligned}
 (4.11) \quad g_{mp+8}(x) - g_{mp+6}(x) &> 0 \quad \text{for } x > x_0 \\
 &= 0 \quad \text{for } x = x_0 \\
 &< 0 \quad \text{for } 0 < x < x_0.
 \end{aligned}$$

It follows that, if $\rho\xi$ is greater than x_0 , the power of T will be a maximum for the linear case [$\therefore \Omega$ has one nonzero root] and a minimum when all the eigenvalues of Ω are equal.

If $\rho\xi$ is less than x_0 , the argument reverses. This argument, which is based on the assumption that the terms of $O((m+n)^{-2})$ are negligibly small, may be illustrated numerically:

Let $p = 3$, $m = 4$ and $n = 204$.

The critical values are:

$$0,1334\rho = 27,7472 \quad \text{at } 1\%$$

$$0,1062\rho = 21,6240 \quad \text{at } 5\%$$

Let $\text{tr}(\Omega) = 6$, then the mode of $\chi^2(18,6)$, x_0 , is approximately 24.

Then, using (4.7), the power of T is approximately:

	<u>EQUAL ROOTS</u>	<u>SINGLE ROOT</u>
1%	0,1153	0,1155
5%	0,2876	0,2873

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